

HRMS & ICMS — Complete Functional Documentation

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****Application Name:**** Innand / Virsme HR Platform

****Classification:**** Internal — Confidential

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1. Executive Summary

This application is a multi-tenant, cloud-hosted **Human Resource Management System (HRMS)** combined with an **Internal Communication Management System (ICMS)**. It is deployed under the brand names **Innand** and **Virsme** and serves small-to-medium enterprises needing end-to-end workforce management.

Business Problems Solved

| Problem | Solution Provided |

|---|---|

| Manual attendance tracking | Real-time digital clock-in/out with session tracking, deduction automation, and audit logs |

| Payroll calculation errors | Automated, configurable, encrypted salary slip generation with tax, PF, gratuity, and loan deductions |

| Leave management chaos | Self-service leave applications with multi-level approval workflows and balance tracking |

| Scattered HR documents | Centralised document generation (offer letters, certificates, NDAs) with digital signatures |

| Lack of employee visibility | Employee self-service portal for salary, leave, attendance, and requests |

| Communication fragmentation | Unified inbox integrating email (IMAP), WhatsApp, internal chat, and threaded discussions |

| Untracked client communications | Assignment message system with approval chains for client-facing work |

| Compliance and data security | AES-256 field encryption for salary data, JWT authentication, 2FA, and full audit trails |

| Onboarding inefficiency | Self-service onboarding link for new hires with step-by-step profile completion |

| Financial request management | Centralised loan, advance salary, reimbursement, and commission request workflows |

2. Application Architecture Summary

Technology Stack

| Layer | Technology |

---|---

| Backend Runtime | Node.js (Express.js) |

| Database | MongoDB (Mongoose ODM) |

| Real-time Communication | Socket.IO |

| Frontend (Admin) | React + TypeScript + Tailwind CSS + Shadcn/ui |

| Frontend (Employee) | React + TypeScript + Tailwind CSS + Shadcn/ui |

| Frontend (Attendance) | React + TypeScript + Tailwind CSS |

| Frontend (Onboarding) | React (complete-profile) |

| Frontend (Connect) | React (communication module) |

| Authentication | JWT (Access + Refresh tokens), bcryptjs, TOTP 2FA |

| Encryption | AES-256-CBC (field-level encryption for salary data) |

| PDF Generation | Puppeteer (headless Chromium) |

| Email Integration | IMAP watcher + SMTP via Nodemailer |

| Scheduled Jobs | node-cron |

| File Uploads | Multer |

| AI Integration | OpenAI / DeepSeek API |

Deployment Architecture

...

Production Domains:

admin.virsme.com → Admin Dashboard (React)

admin.innand.com → Admin Dashboard (alternate brand)

employee.virsme.com → Employee Portal (React)

hr.virsme.com → HR Portal

attendance.virsme.com → Attendance Web App

complete-profile.virsme.com → Onboarding Form

apis.innand.com → Backend API Server (Node.js/Express)

Server:

HTTPS on port 443 (Let's Encrypt TLS)

HTTP on port 4000 (redirect / fallback)

Static files served from /uploads

...

Repository Structure

...

Full stack/

```
├── backend/      ← Node.js/Express API server
|   └── src/
|       ├── index.js    ← Main entry point, route registration, Socket.IO
|       ├── models/     ← 90+ MongoDB Mongoose schemas
|       ├── routes/     ← 80+ Express routers
|       ├── controllers/ ← 89 business-logic handlers
|       ├── middleware/  ← 11 custom middleware (auth, upload, etc.)
|       ├── services/    ← Reusable service classes
|       ├── utils/       ← Encryption, leave calculation, audit logging
|       └── schedulers/  ← Cron jobs (session timeout, leave sync)
```

```
|   └─ watcher.js      ← IMAP email watcher
|
|─ admin_dashboard_hr/  ← Admin / HR React app (50+ pages)
|─ Employee_dashboard/  ← Employee React portal (30+ pages)
|─ attendance/          ← Attendance-only React app
|─ complete-profile/    ← New-hire onboarding form
|─ Connect/             ← Internal communication module
└─ scripts/             ← Database migration scripts
...
---
```

3. Module-wise Functional Documentation

3.1 Authentication & Session Management

****Module Purpose:**** Controls who can access the system, verifies identity, manages active sessions, and enforces automatic logout on inactivity.

Sub-modules

3.1.1 Admin / HR Authentication

****File:**** `backend/src/routes/auth.js`, `backend/src/controllers/authController.js`

| Function | Description |

|---|---|

| ****Register**** (`POST /api/auth/signup`) | Creates a new admin/HR user account. Hashes password with bcryptjs. Stores role, owner reference, and optional 2FA secret. |

| ****Login**** (`POST /api/auth/login`) | Validates credentials. If 2FA is enabled, returns a temporary token and requests OTP. Otherwise returns access + refresh JWT. |

| ****2FA Verify**** (`POST /api/auth/verify-otp`) | Validates TOTP code generated by authenticator app. Issues full JWT on success. |

| ****Setup 2FA**** (`POST /api/auth/setup-2fa`) | Generates a TOTP secret and QR code (via `speakeasy` + `qrcode`). Admin scans this with their authenticator app. |

| ****Token Refresh**** (`POST /api/auth/refresh`) | Accepts a valid refresh token and issues a new access token. Rotation strategy prevents replay attacks. |

| ****Logout**** (`POST /api/auth/logout`) | Invalidates the session. |

| ****Get Profile**** (`GET /api/auth/profile`) | Returns the authenticated admin/HR user's profile. |

****Access:**** Admin, HR, Super-Admin only

****Security:**** JWT verification via `middleware/auth.js`

3.1.2 Employee Authentication

****File:**** `backend/src/routes/empAuth.js`, `backend/src/controllers/empAuthController.js`

| Function | Description |

|---|---|

| **Employee Login** (`POST /api/emp-auth/login`) | Validates employee credentials. Creates an `EmployeeSession` record. Returns access + refresh token. Records device fingerprint, IP, and user agent. |

| **Heartbeat** (`POST /api/emp-auth/heartbeat`) | Called every 30 seconds by the Employee Dashboard. Updates `lastActivity` on the `EmployeeSession` to keep session alive. |

| **Session Reactivation** (`POST /api/emp-auth/reactivate-session`) | When an employee refreshes the browser page, this endpoint marks the session as actively resumed (prevents false half-day deduction). |

| **Logout** (`POST /api/emp-auth/logout`) | Sets session state to `logout`, marks end time. |

| **Token Refresh** (`POST /api/emp-auth/refresh`) | Rotates employee tokens. |

| **Get My Profile** (`GET /api/emp-auth/profile`) | Returns logged-in employee's own record. |

Session Tracking Fields (EmployeeSession model):

- `employeeId` — reference to employee
- `sessionId` — unique identifier
- `loginTime`, `lastActivity`, `logoutTime`
- `sessionState` — `active`, `timeout`, `logout`
- `deviceFingerprint`, `ipAddress`, `userAgent`
- `autoLogoutFlag` — set when system auto-logs out

Token Expiry:

- Employee access token: 16 hours
- Admin access token: 30 minutes (configurable)
- Refresh token: 7 days

3.1.3 HR-Specific Login

****File:**** `backend/src/routes/hrAuth.js`

A separate authentication endpoint for HR staff that may carry different permission scopes than full Admin.

3.1.4 Session Timeout Automation

****File:**** `backend/src/schedulers/sessionTimeoutCron.js`

- Runs on a scheduled interval (every minute)
- Queries all `EmployeeSession` records with `sessionState = active`
- Checks if `lastActivity` is older than 3 minutes
- If stale: sets `sessionState = timeout`, records `logoutTime`, sets `autoLogoutFlag = true`
- This triggers an attendance half-day deduction if the employee did not properly check out

****Business Rule:**** If an employee closes their browser or is idle for 3+ minutes without logging out, the system treats this as an unplanned departure and may apply a half-day deduction.

3.1.5 Middleware (Auth Guards)

****File:**** `backend/src/middleware/`

Middleware Purpose

--- ---

`auth.js`	Verifies JWT for Admin/HR requests
`empAuth.js`	Verifies JWT for Employee requests
`unifiedAuth.js`	Accepts tokens from both User and Employee models
`anyPayrollAuth.js`	Payroll access; supports delegation (HR can act on behalf of owner)
`attendanceAuth.js`	Validates attendance app-specific tokens
`anyAuth.js`	General fallback auth
`payrollAuth.js`	Strict payroll-only access control
`permissions.js`	Role-based feature-level permission check
`autoTaxMiddleware.js`	Automatically recalculates tax when payroll data changes

3.2 Employee Management (HRMS Core)

****Module Purpose:**** Complete employee lifecycle management from creation to offboarding. Stores all personal, employment, and financial profile data.

****File:**** `backend/src/routes/employees.js`, `backend/src/controllers/employeeController.js`

****Model:**** `backend/src/models/Employees.js`

3.2.1 Employee Profile Fields

| Category | Fields |

|---|---|

| ****Personal**** | Full name, Date of birth, Gender, CNIC number (encrypted), Phone, Address, Profile photo |

| ****Employment**** | Employee ID, Department, Sub-department, Designation, Joining date, Employee type (permanent/contract) |

| ****Shift**** | Assigned shift reference, shift start/end times |

| ****Salary**** | Gross salary, salary structure reference, salary effective date |

| ****Bank**** | Bank name, account number, branch code (for salary disbursement) |

| ****Nominee**** | Nominee name, relationship, CNIC, contact |

| ****Status**** | `active`, `pending`, `offered`, `onboarding`, `resigned`, `terminated`, `in\_trash` |

| ****Leave**** | Paid leave quota, unpaid leave allowance, bonus hours |

| ****Probation**** | Probation start/end dates, probation status |

| ****Supervision**** | Supervision mode (`direct` or `needs\_approval`), direct manager reference |

| ****Devices**** | Trusted device list for multi-device control |

| ****Position History**** | Array of past designations and salary changes |

3.2.2 Employee CRUD Operations

Operation	Endpoint	Who Can Access	Notes
List all employees	`GET /api/employees`	Admin, HR	Supports filter by department, status, designation
Create employee	`POST /api/employees`	Admin, HR	Status starts as `pending` or `onboarding`
View employee	`GET /api/employees/:id`	Admin, HR	Returns full profile with nested relations
Update employee	`PUT /api/employees/:id`	Admin, HR	Creates a `ProfileRevision` record for audit
Soft delete	`DELETE /api/employees/:id`	Admin	Moves to `in_trash` — not permanently deleted

| Lightweight list | `GET /api/employees/list` | Admin, HR, some Employees | For dropdowns — only name, ID, department |

| Org chart | `GET /api/employees/hierarchy` | Admin, HR | Returns nested hierarchy for org chart rendering |

| Bulk resign | `POST /api/employees/bulk/resign` | Admin | Sets selected employees to `resigned` |

| Bulk activate | `POST /api/employees/bulk/activate` | Admin | Reactivates employees from trash/resigned |

| Auto-offboard | `GET /api/employees/auto-offboard` | System (cron) | Checks resignation dates and auto-terminates |

3.2.3 Employee Permissions

****File:**** `backend/src/controllers/employeeController.js`

Each employee can have individual permission overrides:

- `GET /api/employees/permissions/:id` — retrieve current permissions
- `PUT /api/employees/permissions/:id` — update permissions

These permissions control what the employee can see or do in the Employee Portal (e.g., view salary slip, view leave balance, access chat).

3.2.4 Profile Revisions

****File:**** `backend/src/routes/profileRevisions.js`, Model: `ProfileRevision`

Every time an employee's profile is edited, the old values are stored in a `ProfileRevision` record. This provides a full audit trail of all profile changes.

- Accessible by Admin/HR via `GET /api/admin/profile-revisions`
- Accessible by the employee themselves via `GET /api/emp-profile-revisions`

3.2.5 Promotions

****File:**** `backend/src/routes/promotion.js`, `backend/src/controllers/promotionController.js`

****Model:**** `Promotion` (implied from routes)

Function	Description
---	---
Create promotion	Records new designation, new salary, and effective date
Auto-calculate new salary	Takes a percentage increase and recalculates all salary components based on the structure
Update employee record	Updates employee's current designation and gross salary
View promotion history	Lists all promotions for an employee

****Access:**** Admin only

3.2.6 Employee Birthdays & Anniversaries

****File:**** `backend/src/routes/empBirthdayRoutes.js`,
`backend/src/routes/anniversariesRoute.js`

- Returns upcoming employee birthdays and work anniversaries
- Used by both Admin Dashboard and Employee Portal for celebration display
- Triggers notifications to admin and peers

3.3 Attendance Management

****Module Purpose:**** Track when employees start and stop work each day, detect tardiness and early departure, calculate deductions, and maintain a tamper-proof audit trail.

****File:**** `backend/src/routes/attendance.js`, `backend/src/controllers/attendanceController.js`

****Model:**** `backend/src/models/Attendance.js`, `AttendanceLog.js`, `EmployeeSession.js`

3.3.1 Attendance Check-In / Check-Out

****Endpoint:**** `POST /api/attendance`

****Process:****

1. Employee presses the Check-In button on the attendance app or portal
2. System records timestamp, session ID, IP, device fingerprint, and user agent
3. System compares clock-in time against the employee's assigned shift start time
4. If clock-in is after `shift\_start + grace\_period` → marks as ****Late****
5. On Check-Out:
 - Calculates total working hours
 - If checkout is before threshold (e.g., 6 PM) → marks as ****Half-Day****
 - Calculates deductions accordingly

****Status Values:****

- `present` — Normal attendance
- `absent` — No clock-in recorded
- `half\_day` — Clocked out before the threshold time

- `late` — Clocked in after grace period
- `on\_leave` — Leave was approved for this date
- `holiday` — Date is a non-working day

3.3.2 Attendance Configuration

****File:**** `backend/src/routes/attendanceConfig.js`

****Model:**** `AttendanceConfig`

Configurable rules:

- `graceMinutes` — Minutes after shift start before marking late
- `halfDayThreshold` — Cutoff time for marking half-day
- `workingDaysPerWeek` — Work schedule
- `lateDeductionRules` — How much to deduct per late occurrence

3.3.3 Attendance Challenges

****File:**** Part of `backend/src/routes/attendance.js`

****Model:**** `AttendanceChallenge`

If an employee believes their attendance record is incorrect, they can raise a challenge:

1. Employee submits: `PATCH /api/attendance/:id/challenge`
2. Admin reviews the challenge via the challenges list: `GET /api/attendance/challenges`
3. Admin approves or rejects with notes
4. If approved, the attendance record is corrected and the audit log is updated

****Important Note:**** There is an identified bug in the original system where checkIn/checkOut fields were not updating correctly when a challenge was approved. This has been flagged as a known issue (see section 11).

3.3.4 Attendance Logs & Audit Trail

****File:**** `backend/src/routes/attendance.js` (logs endpoints)

****Model:**** `AttendanceLog`, `AttendanceChangeLog`

- `GET /api/attendance/audit-logs` — Full history of all manual corrections
- `GET /api/attendance/logs/between-shift` — Tracks logins that occur between two shifts (unusual logins)
- `PATCH /api/attendance/logs/between-shift/:id` — Admin can flag or resolve between-shift logs

****AttendanceChangeLog fields:**** employeeId, date, changedBy (admin user ID), oldValue, newValue, timestamp

3.3.5 Browser Close / Refresh Detection

****Files:**** `BROWSER\_CLOSE\_DETECTION\_GUIDE.md`, `REFRESH\_DETECTION\_FIX.md`

The system distinguishes between:

- ****Browser close (intentional departure):**** Session heartbeat stops; after 3 minutes timeout → half-day deduction applied
- ****Page refresh (temporary interruption):**** Session reactivation endpoint is called immediately on page reload → no deduction

This is handled via the `sessionTimeoutCron.js` and the `reactivate-session` endpoint.

3.3.6 Between-Shift Login Tracking

****File:**** `BETWEEN\_SHIFT\_LOGIN\_FEATURE.md`

If an employee logs in outside their scheduled shift hours, it is flagged as a "between-shift login." Admin can review, approve (as overtime), or flag it as anomalous.

3.3.7 Manual Date/Attendance Correction

****File:**** `backend/src/routes/setDate.js`

Admin can manually set or correct an attendance record for any past date.

****Endpoint:**** `POST /api/setDate`

****Access:**** Admin, HR only

****Creates:**** `AttendanceChangeLog` record for audit

3.3.8 Non-Working Days / Holidays

****Model:**** `SpecificNonWorkingDay`

****Route:**** `/api/specific-non-working-days`

Admin can define company-specific holidays. The attendance engine uses this to skip deductions on declared holidays.

3.3.9 Attendance Access Control

****Model:**** `AttendanceAccess`

****Route:**** `/api/attendance-access`

Controls which admin users or roles can view or modify attendance records for specific departments.

3.4 Leave Management

****Module Purpose:**** Allow employees to apply for leave, manage leave balances, process approvals, and link leave data to payroll deductions.

****Files:**** `backend/src/routes/leaves.js`, `backend/src/routes/applyLeaveRoutes.js`,
`backend/src/routes/employeeLeaves.js`

****Models:**** `Leaves`, `LeaveRecord`, `LeaveTransaction`, `LeaveYearBalance`, `ApplyLeave`

3.4.1 Leave Types

****Model:**** `Leaves`

Admin defines available leave types:

- Annual Leave
- Sick Leave
- Casual Leave
- Unpaid Leave
- Custom types (configurable)

Each type has:

- Maximum days per year
- Carryforward rules (can unused days roll to next year?)
- Paid / unpaid flag

3.4.2 Leave Application (Employee)

****Endpoint:**** `POST /api/apply-leave` or `POST /api/emp-leaves/apply`

****Process:****

1. Employee selects leave type, start date, end date, and reason
2. System checks available leave balance for the selected type
3. System checks for conflicts (existing approved leaves, holidays)
4. Application is submitted with status `pending`
5. Manager/HR is notified in real-time via Socket.IO
6. Leave is not deducted until approved

3.4.3 Leave Approval (Admin / HR)

****Endpoints:****

- `PATCH /api/emp-leaves/:id/approve` — Approve leave
- `PATCH /api/emp-leaves/:id/reject` — Reject leave with reason

****On Approval:****

1. Leave status updated to `approved`
2. `LeaveTransaction` record created (debit from balance)
3. `LeaveYearBalance` updated
4. Attendance record for those dates updated to `on\_leave`
5. Employee notified via Socket.IO

3.4.4 Leave Balance

****Endpoint:**** `GET /api/leaves/balance/:employeeId`

****Model:**** `LeaveYearBalance`

Returns remaining balance per leave type for the current leave year. Calculated from entitlement minus approved transactions.

3.4.5 Probation Leave

****Model:**** `ProbationLeaveApproval`

****Route:**** `/api/probation-leave-approvals`

Employees on probation have restricted leave entitlements. A separate approval flow applies for leaves requested during the probation period.

3.4.6 Leave Encashment

****Model:**** `LeaveEncashmentRequest`

****Route:**** `/api/leave-encashment-requests`

Employee can request to convert unused leave balance into cash payment:

1. Employee submits encashment request with number of days
2. HR reviews and approves
3. Amount is added to the next salary slip
4. Leave balance is reduced accordingly

3.4.7 Leave Carry Forward

****Model:**** `LeaveCarryForwardRequest`

****Route:**** `/api/leave-carry-forward-requests`

At year end, employee can request to carry unused leave to the next year (subject to maximum carryforward limits).

3.4.8 Leave Sync Cron

****File:**** `backend/src/schedulers/leaveSyncCron.js`

Runs nightly to:

- Reset annual leave balances at the start of a new leave year
- Process carry-forward requests
- Sync leave balances with attendance records

3.5 Payroll & Salary Management

****Module Purpose:**** Automate monthly payroll calculation including salary components, deductions, tax, PF, gratuity, and loan EMIs. Generate encrypted salary slips and email them to employees.

****Files:**** `backend/src/routes/salarySlips.js`, `backend/src/routes/salarySettings.js`,
`backend/src/routes/salaryStructure.js`

****Models:**** `SalarySlip`, `SalaryStructure`, `SalarySettings`, `PayrollPeriod`

3.5.1 Salary Structure Configuration

****Model:**** `SalaryStructure`

****Route:**** `/api/salary-structure`

Defines percentage-based breakdown of gross salary:

- ****Basic Salary**** — e.g., 60% of gross
- ****HRA (House Rent Allowance)**** — e.g., 20%

- **Medical Allowance** — e.g., 7%
- **Transport Allowance** — e.g., 5%
- **Utility Allowance** — e.g., 4%
- **Other/Custom Components** — configurable

Admin configures these percentages once; all salary slips use this structure.

3.5.2 Salary Settings

Model: `SalarySettings`

Route: `/api/salary-settings`

Setting	Description
---	---
`workingDaysPerMonth`	Standard working days (e.g., 26)
`payrollCutoffDate`	Day of month when payroll is locked
`lateDeductionFormula`	How to calculate late-arrival deductions
`halfDayDeductionRate`	Percentage of daily rate deducted for half-day
`taxBrackets`	Income tax slabs configuration
`pfRate`	Provident Fund contribution percentage
`gratuitySettings`	End-of-service gratuity formula

3.5.3 Payroll Periods

Model: `PayrollPeriod`

Route: `/api/payroll-periods`

Each calendar month has a payroll period:

- `startDate`, `endDate`
- `status` — `open`, `locked`
- `isLocked` — When locked, no further edits to slips for that month

****Admin locks the period**** after reviewing all slips to prevent post-approval changes.

3.5.4 Salary Slip Generation

****Endpoint:**** `POST /api/salary-slips/generate`

****Access:**** Admin, HR (via `anyPayrollAuth`)

****Generation Process:****

1. Admin selects payroll period
2. System fetches all active employees
3. For each employee:
 - a. Get gross salary and structure percentages
 - b. Calculate component amounts (Basic, HRA, etc.)
 - c. Fetch attendance data → calculate working days
 - d. Apply late deductions (from `AttendanceConfig`)
 - e. Apply half-day deductions
 - f. Deduct approved unpaid leaves
 - g. Add loan EMI deductions (from `LoanDetail`)
 - h. Calculate tax using bracket table
 - i. Calculate PF deduction
 - j. Add any scheduled allowances (bonuses, commissions)
 - k. Calculate net salary
4. Each slip is ****encrypted using AES-256-CBC**** before storage

5. Slip stored in `SalarySlip` collection with `encryptedData` field

3.5.5 Salary Slip Encryption

****File:**** `backend/src/utils/crypto.js`, `backend/src/utils/encryption.js`

****Model:**** `DecryptionKey`

- All salary data is encrypted at rest using AES-256-CBC
- A master decryption key is stored in the `DecryptionKey` model (accessible only to authorised admin)
- To view a slip, admin must provide the master key: `GET /api/salary-slips/decrypt/:id`
- Employee can view their own slip via Employee Portal (decrypted on-the-fly with server-side key)

3.5.6 Manual Salary Slips

****Route:**** `/api/manual-slip`

For situations where the automated generation is insufficient (e.g., mid-month joins, bonuses), admin can create an ad-hoc salary slip with a visual grid editor.

3.5.7 Send Salary Slip by Email

****Endpoint:**** `POST /api/send-slip-email/:id`

Generates a PDF version of the salary slip using Puppeteer and emails it to the employee's registered email address via SMTP.

3.5.8 Salary Change Requests

****Model:**** `SalaryChangeRequest`

****Route:**** `/api/salary-change-requests`

Function	Description
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Employee/Manager submits request	States current salary, requested new salary, reason
----------------------------------	---

HR reviews	Can approve, reject, or request more information
------------	--

On approval	Employee's gross salary is updated; position history is recorded
-------------	--

3.5.9 Custom Salary Slip Fields

****Route:**** `/api/salary-fields`

****Model:**** `SalarySlipFields` (implied)

Admin can define additional custom fields that appear on salary slips (e.g., performance bonus, project incentive).

3.5.10 Payroll Estimates

****Route:**** `/api/payroll-estimates`

****Model:**** `PayrollEstimate`

Admin can generate a preview/estimate of next month's payroll before finalising it. Useful for budget planning.

3.6 Financial Requests & Benefits

3.6.1 Loan Management

****Route:**** `/api/loans`, `/api/loan-requests`

****Model:**** `LoanRequest`, `LoanDetail`

| Step | Description |

|---|---|

| Application | Employee applies via `POST /api/loan-requests` with amount and reason |

| Review | HR/Admin reviews via loan requests list |

| Approval | Admin sets EMI count, interest rate, deduction start date |

| EMI Schedule | System generates monthly deduction schedule in `LoanDetail` |

| Payroll deduction | Each month, salary slip automatically deducts the EMI |

| Settlement | Loan marked as settled when all EMIs are paid |

****Loan Calculator**** (Admin Dashboard screen): Standalone tool to calculate interest and generate amortisation schedule before approving.

3.6.2 Advance Salary Requests

****Route:**** `/api/advance-salary-requests`

****Model:**** `AdvanceSalaryRequest`

Employee requests early payment of next month's salary. On approval, amount is added to current payroll and deducted from next month.

3.6.3 Reimbursement Requests

****Route:**** `/api/reimbursement-requests`

****Model:**** `ReimbursementRequest`

| Field | Description |

|---|---|

| Expense category | Travel, food, equipment, etc. |

| Amount | Claimed amount |

| Receipt | Uploaded proof |

| Approval status | pending, approved, rejected |

| Payroll link | Approved amount added to next salary slip |

3.6.4 Bonus Requests

****Route:**** `/api/bonus-requests` (inferred)

****Model:**** `BonusRequest`

Manager or HR can initiate a bonus for an employee. Goes through approval before being added to payroll.

3.6.5 Commission Requests

****Route:**** `/api/commission-requests`

****Model:**** `CommissionRequest`

Sales team members can submit commission claims based on deals closed. HR verifies and adds to payroll.

3.6.6 Provident Fund (PF)

****Route:**** `/api/pf`

****File:**** `backend/src/routes/pf.js`

- Calculates employee and employer PF contributions
- Tracks cumulative PF balance per employee

- Accessible to admin for reporting and employee for personal view

3.6.7 Gratuity

****Route:**** `/api/gratuity`

****Model:**** `GratuitySetting`

- Configured formula based on years of service and last drawn salary
- Admin can view estimated gratuity liability per employee
- Calculated at time of resignation/termination for final settlement

3.6.8 Tax Management

****Route:**** `/api/tax`

****Models:**** `TaxStatus`, `TaxOverride`, `TaxAdjustmentRequest`

Function	Description
----------	-------------

---	---
-----	-----

Tax brackets	Admin defines income tax slabs (`POST /api/tax/brackets`)
--------------	---

Employee tax status	View current tax category for each employee
---------------------	---

Tax override	Manually override tax calculation for an employee
--------------	---

Tax adjustment requests	Employee can request a tax re-evaluation; HR reviews
-------------------------	--

3.6.9 Unified Financial Requests View

****Route:**** `/api/unified-requests`

Single endpoint that returns all types of financial requests (loans, advances, reimbursements, bonuses, commissions) in one paginated list. Used by the "Payment Requests" screen in Admin Dashboard for bulk review.

3.7 Recruitment & Onboarding

3.7.1 Offer Letter Management

****Route:**** `/api/offer-letter`

****Models:**** `OfferLetter`, `OfferEmailTemplate`, `OfferEmailGenerated`

| Function | Description |

|---|---|

| Template editor | WYSIWYG editor to create/modify offer letter templates |

| Generate offer | Fill in candidate details, populate template |

| Email offer | Send offer PDF to candidate's email |

| Track status | Monitor whether offer was opened, accepted, or declined |

3.7.2 Onboarding

****Route:**** `/api/onboarding`

****App:**** `complete-profile/` (separate React app)

****Onboarding Flow:****

1. Admin creates a placeholder employee record
2. Admin sends a unique onboarding link to the new hire's email
3. New hire opens the link and completes the `complete-profile` form:
 - Personal information
 - Emergency contacts

- Bank account details
- CNIC upload
- Photograph upload

4. Submitted data populates the employee record in the database

5. Admin reviews and approves the profile

6. Employee status changes from `onboarding` to `active`

7. System generates login credentials for the employee portal

****Manual Onboarding:**** Admin can also create an employee record directly without sending a link.

3.7.3 Document Requests for Candidates

****Route:**** `/api/document-requests`

****Model:**** `DocumentRequest`

Admin can request specific documents from candidates (CNIC, CV, certificates) before formally creating their employee profile. The candidate receives a secure upload link.

3.8 Document Management

3.8.1 Document Generation (PDFs)

****Route:**** `/api/generate`

****Technology:**** Puppeteer (headless Chrome)

Document Type	Description
---------------	-------------

|---|---|

| ****Offer Letter**** | Generated from template with candidate details |

| ****NDA**** | Non-disclosure agreement with digital signatures |

| ****Experience Certificate**** | Employment verification document |

| ****Salary Certificate**** | States current salary for bank/visa purposes |

| ****Contract**** | Employment contract with standard clauses |

All PDFs embed digital signatures from HR, CEO, and Employee (`Signature` model).

3.8.2 Employee Documents

****Route:**** `/api/employee-docs`

****Model:**** `EmployeeDoc`

Stores documents uploaded by or for employees (CNIC scans, certificates, passports). Accessible to Admin/HR for compliance purposes.

3.8.3 Document Templates

****Model:**** `DocTemplate`

Admin can create and store reusable document templates for different document types. Templates support dynamic placeholder fields.

3.8.4 Digital Signatures

****Model:**** `Signature`

****Route:**** `/api/signature`

Stores signature images for:

- HR Signature
- CEO/Director Signature
- Employee Signature

These are embedded into generated PDF documents.

3.9 HR Operations & Policy

3.9.1 HR Policies

****Route:**** `/api/hr-policy`

****Model:**** `HrPolicy`

- Rich-text policy editor (HTML content)
- Policies are versioned
- Published policies are visible to employees in the portal
- Categories: Code of Conduct, Leave Policy, Attendance Policy, etc.

3.9.2 HR Rules

****Route:**** `/api/hr-rules`

****Model:**** `HrRules`

System-level HR rules that automate certain behaviours (e.g., automatic warning after N late arrivals).

3.9.3 Warnings

****Route:**** `/api/warnings`

****Models:**** `WarningConfig`, `EmployeeWarning`

Function	Description
----------	-------------

---	---
-----	-----

Warning configuration	Define escalation rules (verbal → written → final)
-----------------------	--

Issue warning	Admin issues a formal warning to an employee with reason and category
---------------	---

Warning history	Full history per employee
-----------------	---------------------------

Employee notification	Employee is notified of the warning via the portal
-----------------------	--

3.9.4 Penalties

****Route:**** `/api/penalties`

****Model:**** `Penalty`

Financial or administrative penalties issued to employees. Penalty amounts can be linked to payroll for automatic deduction.

3.9.5 Probation Periods

****Route:**** `/api/probation-periods`

****Model:**** `ProbationPeriod`

Function	Description
----------	-------------

---	---
-----	-----

Create probation	Set start/end date for a new employee
------------------	---------------------------------------

Track status	Monitor if probation is ongoing, extended, or completed
--------------	---

| Special leave rules | During probation, leave is restricted (handled by
`ProbationLeaveApproval`) |

| Completion | Admin marks probation as passed or extended |

3.9.6 Notice Period

****Route:**** `/api/notice-period`

****Model:**** `NoticePeriod`

When an employee resigns:

1. Notice period duration is configured per designation
2. Last working date is calculated
3. Final settlement is initiated
4. Employee's status transitions to `resigned` on the last working day

3.9.7 Scheduled Allowances

****Route:**** `/api/payroll-schedule` (scheduled allowances)

****Model:**** `ScheduledAllowance`

Admin can set recurring allowances (e.g., transport allowance every month, annual bonus in December) that are automatically added to payroll.

3.10 Communication Hub (ICMS)

****Module Purpose:**** Centralise all internal and external communication into one platform — email, WhatsApp, internal chat, and threaded discussions.

3.10.1 Internal Chat

****Route:**** `/api/chat`

****Models:**** `Chat`, `ChatThread`

****Technology:**** Socket.IO real-time

Function Description

--- ---

Send message Real-time message delivery between employees

View history Persistent message history

File attachments Upload and share files in chat

3.10.2 Threaded Chat (Team Discussions)

****Route:**** `/api/thread-chat`

****Model:**** `ThreadChatMessage`

****Technology:**** Socket.IO

Full-featured messaging with:

- ****Threaded replies**** — Reply to specific messages
- ****Emoji reactions**** — Add/remove emoji reactions to messages
- ****Edit messages**** — Modify sent messages (edit history tracked)
- ****Soft delete**** — Delete messages (marked deleted, not removed from DB)
- ****File attachments**** — Documents, images
- ****@ Mentions**** — Tag specific users
- ****Read receipts**** — Mark messages as read
- ****AI summary**** — Generate a summary of thread context using AI

- **Full-text search** — Search within thread messages

Socket.IO Events:

- ``send_thread_message`` — Broadcast new message to all room members
- ``edit_thread_message`` — Notify of edits
- ``delete_thread_message`` — Remove from view
- ``add_thread_reaction`` / ``remove_thread_reaction`` — Reaction updates
- ``mark_thread_read`` — Read status sync
- ``get_thread_summary`` — AI-powered summary request

3.10.3 Email Client (IMAP Integration)

File: ``backend/src/watcher.js``, ``backend/src/routes/emailReceiverRoutes.js``

Models: ``EmailTemplate``, ``EmailLabel``, ``EmailSignature``

Fully integrated email client:

- **IMAP watcher** polls for new incoming emails in real-time
- **Compose and send** via SMTP (Nodemailer)
- **Labels/folders** for email organisation
- **Email signatures** per user
- **Templates** for common responses
- **Thread view** — Emails are grouped by conversation
- **Attachments** — Full support for sending and viewing attachments
- **Unread count** — Tracked and displayed in notification badge
- **Email filters** — Filter by label, sender, date

Route: ``/api/email-receiver``

3.10.4 WhatsApp Integration

****Route:**** `/api/whatsapp-messages`

****Model:**** `WhatsAppMessage`

****Technology:**** 360Dialog or Twilio API (needs confirmation)

- Send and receive WhatsApp messages from within the platform
- Messages are stored in the database
- Accessible from the employee communication hub

3.10.5 Chat Thread Routes

****Route:**** `/api/chat-threads`

****Model:**** `ChatThread`

Organisational layer for grouping multiple chats into threads/topics for better management.

3.11 Client & Assignment Management

****Module Purpose:**** Manage client relationships and coordinate internal team communication regarding client work through an approval-based message system.

3.11.1 Client Information

****Route:**** `/api/client-info`

****Model:**** `ClientInfo`

Function	Description
---	---
Create client	Add client/account with company name, contacts, and industry
Update client	Modify client details
View clients	List all clients with assigned employees
Delete client	Soft delete a client record

3.11.2 Assignment Messages

****Route:**** `/api/assignment-messages`

****Model:**** `AssignmentMessage`

The core ICMS workflow for client-related communications:

****Workflow:****

1. A team member composes a message related to a client
2. The message is sent to an approval chain (managers, seniors)
3. Each approver in the chain reviews and approves or disapproves
4. Only approved messages are considered official/sent
5. Resubmission is allowed after disapproval

****Message Fields:****

- `sender` — Who composed the message
- `receivers` — Target recipients
- `clientReference` — Linked client
- `content` — Message body

- `attachments` — File attachments
- `approvalChain` — Array of approvers and their decisions
- `status` — `pending`, `approved`, `disapproved`

****Socket.IO Events:****

- `send\_assignment\_message` — New message broadcast
- `assignment\_message\_approved` / `disapproved` — Status updates
- `assignment\_message\_edited` — Edit notifications
- `assignment\_message\_resubmitted` — After disapproval correction

****Access:****

- Employees: Create and view their own messages
- Managers: Approve/disapprove messages in their chain
- Admin: Full access to all messages

3.12 Task & Workspace Management

****Route:**** `/api/tasks`, `/api/admin/task`, `/api/admin-workspace-management`,
 `/api/employee-task`

****Models:**** `Task`, `TaskSpace`, `Workspace`

Function	Description
---	---
Create task	Admin or manager creates a task with title, description, assignees, and due date

- | Assign task | Assign to one or multiple employees |
- | Update status | Employee updates task status (todo, in-progress, done) |
- | View tasks | Employee sees their assigned tasks; Admin sees all |
- | Workspace organisation | Tasks grouped into workspaces for project-level organisation |

****ClickUp Integration:**** The system has routes/components referencing ClickUp for task synchronisation, allowing tasks from ClickUp to be visible within the employee portal.

3.13 Organisational Structure

3.13.1 Departments

****Route:**** `/api/departments`

****Model:**** `Departments`

- Create/edit/delete departments
- Each department has a name and optionally a head (manager reference)
- Employee count is calculated dynamically

3.13.2 Sub-Departments

****Route:**** `/api/subdepartments`

****Model:**** `SubDepartments`

- Nested under departments for fine-grained organisational structure
- Employees can belong to a sub-department

3.13.3 Designations

****Route:**** `/api/designations`

****Model:**** `Designations`

- Job title definitions
- Optional salary grade assignment
- Career level tracking (junior, senior, lead, manager, director)

3.13.4 Hierarchy / Org Chart

****Route:**** `/api/org-hierarchy`, `/api/hierarchy`

****Models:**** `EmployeeHierarchy`, `OrgHierarchy`

- Builds and retrieves the full organisational chart
- Visualises reporting lines
- Used by the admin dashboard's Hierarchy page for org chart rendering

3.14 Settings & System Configuration

3.14.1 System Settings

****Route:**** `/api/settings`

****Model:**** `Settings`

Global application settings including timezone, locale, and system-level preferences.

3.14.2 Font Settings

****Route:**** `/api/font-setting`

****Model:**** `FontSetting`

Allows admin to change the UI font family across the platform (Google Fonts integration).

3.14.3 Company Profile

****Route:**** `/api/company-profile`

****Model:**** `CompanyProfile`

Company name, logo, address, registered number, and HR contact details used in document headers and email footers.

3.14.4 Role-Based Access Control

****Route:**** `/api/role`

****Model:**** `RolePermission`

- Define custom roles with specific feature permissions
- Each permission maps to a UI feature or API action
- Admin can create custom roles beyond the built-in ones

3.14.5 User Management

****Route:**** `/api/users`

****Model:**** `Users`

- Create and manage HR/Admin user accounts
- Assign roles
- Enable/disable 2FA
- Reset passwords

3.14.6 Extra / Custom Fields

****Route:**** `/api/extra-fields`

****Model:**** `Extrafields`

Admin can define custom data fields that appear on employee profiles. Useful for company-specific data not covered by default fields.

3.14.7 Decryption Keys

****Route:**** `/api/decryption-keys`

****Model:**** `DecryptionKey`

Master keys for decrypting salary slip data. Access is highly restricted (Admin only). Multiple keys supported for key rotation.

3.15 Reports, Analytics & Dashboard

3.15.1 Admin Dashboard Overview

****Screen:**** Admin Dashboard Home

Live statistics displayed:

- Total active employees
- Monthly payroll total
- Monthly attendance rate (%)
- Pending leave requests count
- Upcoming probation expirations
- Recent activity feed

3.15.2 Payroll Summary

****Screen:**** Payroll Module

- Monthly payroll total cost (gross + employer contributions)
- Net payroll (after all deductions)
- Total deductions breakdown
- Employee-wise salary breakdown table
- Exportable to Excel

3.15.3 Attendance Reports

****Screen:**** Attendance Module

- Daily attendance grid (all employees, colour-coded by status)
- Monthly attendance stats per employee
- Department-wise attendance summary
- Late arrival frequency report
- Absent employees list

3.15.4 Leave Reports

****Screen:**** Leave Dashboard

- Calendar view of all leaves across the company
- Leave balance per employee per type
- Pending vs. approved vs. rejected breakdown
- Department-wise leave utilisation

3.15.5 Financial Reports

****Screen:**** Payment Requests, Loans, Tax Manager

- Loan EMI schedule per employee
- PF balance and contributions
- Gratuity liability estimate
- Tax deduction summary per employee

4. Role and Permission Matrix

Feature	Super-Admin	Admin	HR	Manager	Employee
--- --- --- --- --- ---					
Employee Management					
Create employee	✓	✓	✓	X	X
View all employees	✓	✓	✓	Dept only	Own only
Edit employee profile	✓	✓	✓	X	Own (limited)
Delete employee	✓	✓	X	X	X

Manage permissions	✓	✓	X	X	X	
Attendance						
Check-in / Check-out	X	X	X	✓	✓	
View all attendance	✓	✓	✓	Dept only	Own only	
Manually correct attendance	✓	✓	✓	X	X	
Approve challenges	✓	✓	✓	X	X	
Configure rules	✓	✓	X	X	X	
Leave						
Apply for leave	X	X	✓	✓	✓	
Approve / Reject leave	✓	✓	✓	✓ (team)	X	
View leave balance	✓	✓	✓	Team	Own only	
Create leave types	✓	✓	✓	X	X	
Payroll						
Generate salary slips	✓	✓	✓	X	X	
View salary slips	✓	✓	✓	X	Own only	
Decrypt salary data	✓	✓	With key	X	Own only	
Configure salary structure	✓	✓	X	X	X	
Lock payroll period	✓	✓	X	X	X	
Email salary slips	✓	✓	✓	X	X	
Financial Requests						
Submit loan request	X	X	X	✓	✓	
Approve loan	✓	✓	✓	X	X	
Submit reimbursement	X	X	X	✓	✓	
Approve reimbursement	✓	✓	✓	X	X	

HR Operations
Issue warning ✓ ✓ ✓ X X
Issue penalty ✓ ✓ ✓ X X
Manage probation ✓ ✓ ✓ X X
Create policies ✓ ✓ ✓ X X
Documents
Generate offer letter ✓ ✓ ✓ X X
Generate certificate ✓ ✓ ✓ X X
View own documents - - - - ✓
Communication
Send internal messages ✓ ✓ ✓ ✓ ✓
Access email client ✓ ✓ ✓ ✓ ✓
Send/receive WhatsApp ✓ ✓ ✓ ✓ ✓
Approve assignment msg ✓ ✓ ✓ ✓ (chain) X
Settings
System settings ✓ ✓ X X X
User management ✓ ✓ X X X
Role management ✓ X X X X
Encryption keys ✓ ✓ X X X

> ****Note:**** `anyPayrollAuth` middleware allows Admin users to delegate payroll access to specific HR accounts. This means some HR operations can be scoped beyond standard HR role permissions.

5. Workflow Documentation

5.1 Employee Onboarding Workflow

...

[Admin] Creates placeholder employee record



[System] Generates unique onboarding link



[Admin] Sends link to candidate's email



[Candidate] Opens complete-profile app



[Candidate] Fills personal details, bank info, uploads CNIC, photo



[System] Saves data to employee record (status: onboarding)



[Admin] Reviews submitted profile



[Admin] Approves → status changes to "active"

[Admin] Rejects → candidate notified to correct and resubmit



[System] Creates employee login credentials



[Employee] Receives credentials and can access Employee Portal

...

5.2 Leave Application Workflow

...

[Employee] Submits leave application

- Selects: leave type, dates, reason



[System] Validates leave balance

- Sufficient balance? → Continue
- Insufficient? → Error shown to employee



[System] Checks for date conflicts (existing leaves, holidays)



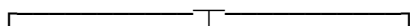
[System] Creates ApplyLeave record (status: pending)



[Socket.IO] Notifies manager / HR in real-time



[Manager/HR] Reviews the request



[Approve]

[Reject]



[System] Updates leave record to "approved" [System] Updates leave record to "rejected"

[System] Creates [Employee] Notified

LeaveTransaction with rejection reason

[System] Deducts balance

[Attendance] Updated

to "on_leave" for dates

[Employee] Notified

...

5.3 Monthly Payroll Workflow

...

[Admin] Opens Payroll Period for current month



[Admin] Clicks "Generate Salary Slips"



[System] For each active employee:

1. Fetch gross salary + salary structure
2. Calculate component amounts (Basic, HRA, etc.)
3. Count working days from attendance
4. Apply late deductions
5. Apply half-day deductions
6. Deduct unpaid leaves
7. Fetch and apply loan EMIs
8. Calculate income tax (bracket-based)
9. Calculate PF deduction
10. Add approved allowances/bonuses
11. Calculate NET salary

12. Encrypt with AES-256

13. Save SalarySlip record



[Admin] Reviews generated slips



[Admin] Edits individual slips if needed (manual adjustments)



[Admin] Sends slips via email (optional bulk send)



[Admin] Locks the Payroll Period



[System] No further edits possible for that period

...

5.4 Attendance Challenge Workflow

...

[Employee] Notices incorrect attendance record



[Employee] Submits challenge via portal

- States reason, correct time

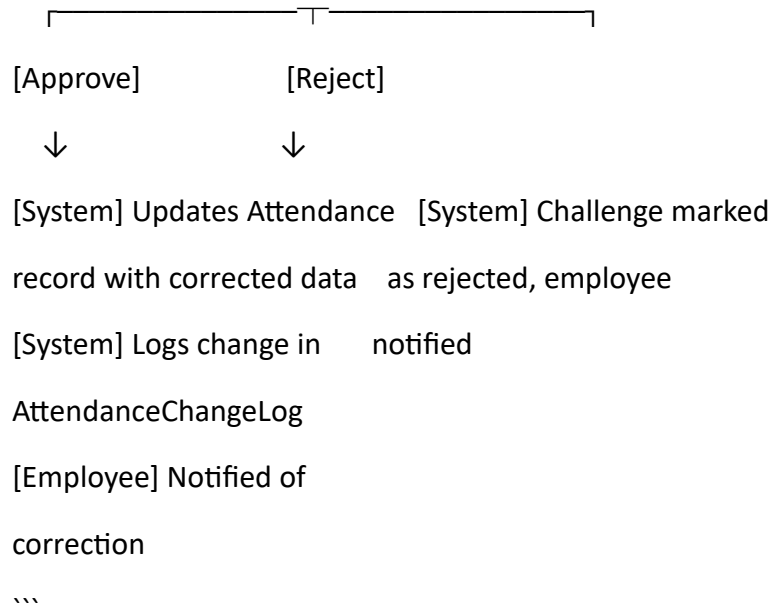


[System] Creates AttendanceChallenge record (status: pending)

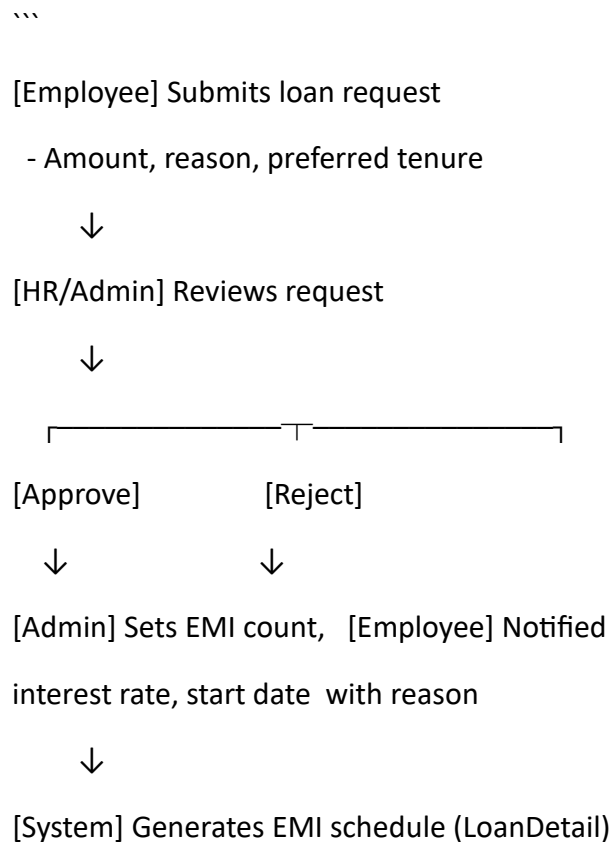


[Admin/HR] Reviews challenges list





5.5 Loan Application Workflow





[Every month during payroll generation]

[System] Deducts current EMI from salary slip



[System] Tracks remaining balance



[Last EMI paid] → Loan marked as settled

...

5.6 Assignment Message (ICMS) Workflow

...

[Employee] Composes client-related message

- Selects client, adds content, attachments



[System] Routes to approval chain



[Approver 1] Reviews message

- Approves → moves to next approver
- Disapproves → returned to sender with comments



[Approver N] Final approval



[Message status → "approved"]



[All relevant parties notified via Socket.IO]

...

5.7 Browser Close Detection Workflow

...

[Employee] Opens Employee Dashboard



[System] Creates EmployeeSession (state: active)



[Every 30 seconds] Heartbeat sent to /api/emp-auth/heartbeat



CASE 1: Employee clicks Logout

[System] Session state → "logout", logoutTime recorded

[No deduction applied]



CASE 2: Employee refreshes page

[Page reload] immediately calls /api/emp-auth/reactivate-session

[System] Session remains active

[No deduction applied]



CASE 3: Employee closes browser

[Heartbeats stop]

[sessionTimeoutCron runs every minute]

[After 3 minutes of no heartbeat → session state → "timeout"]

[autoLogoutFlag → true]

[Half-day deduction may be applied to today's attendance]

...

5.8 Promotion Workflow

...

[Admin] Opens Promotion screen



[Admin] Selects employee, enters:

- New designation
- Salary increase % (or new absolute amount)
- Effective date



[System] Calculates new component amounts based on salary structure



[Admin] Confirms



[System] Creates Promotion record

[System] Updates employee's:

- Current designation
- Gross salary
- Salary effective date

[System] Records old values in position history

...

6. Screen / Page-wise Documentation

6.1 Admin Dashboard — Pages

Screen	Purpose	Key Actions	Data Source
Dashboard Home	Overview of org health	View stats, quick links	Employees, Attendance, PayrollPeriod
Employees List	Browse all employees	Search, filter, bulk actions, add new	Employees collection
Employee Detail	Full employee profile	Edit, view documents, permissions	Employee + related models
Employee Edit	Edit profile form	Save, cancel, upload photo	Employees
Hierarchy	Org chart visual	View reporting lines	OrgHierarchy
Attendance	Daily attendance grid	Filter, manual correction, set date	Attendance
Attendance Settings	Configure rules	Set grace period, half-day threshold	AttendanceConfig
Attendance Challenges	Review disputes	Approve, reject, add notes	AttendanceChallenge
Attendance Logs	Full audit trail	View by date, employee, device	AttendanceLog, AttendanceChangeLog
Leave Dashboard	Company-wide leave view	Calendar, filter, approve/reject	ApplyLeave, LeaveRecord
Leave Types	Manage leave types	Create, edit, delete	Leaves
Payroll	Monthly payroll overview	Generate, review, lock, export	SalarySlip, PayrollPeriod
Salary Slips	All slips listing	View, decrypt, edit, email	SalarySlip
Manual Slip Editor	Ad-hoc slip creation	Visual grid edit, recalculate net	SalarySlip (manual)

| ****Salary Structure**** | Component percentages | Configure breakdown | SalaryStructure |

| ****Salary Settings**** | Global payroll rules | Working days, cutoff date | SalarySettings |

| ****Payroll Estimates**** | Pre-run estimation | View estimates | PayrollEstimate |

| ****Promotion**** | Promote employees | Select, set salary %, confirm | Promotion, Employees |

| ****Loan Calculator**** | Standalone calculator | Enter amount, rate, tenure | (No DB) |

| ****Loan Requests**** | Review loan applications | Approve, reject, set schedule | LoanRequest |

| ****Loan Calculations**** | PF/EOBI estimates | View contributions | PF, EOBI settings |

| ****Advance Salary**** | Review advance requests | Approve, reject | AdvanceSalaryRequest |

| ****Reimbursements**** | Review expense claims | Approve, reject, categorise | ReimbursementRequest |

| ****Commission Requests**** | Review commissions | Approve, verify | CommissionRequest |

| ****Bonus Requests**** | Review bonuses | Approve | BonusRequest |

| ****Salary Change Requests**** | Review salary changes | Approve, reject | SalaryChangeRequest |

| ****Tax Manager**** | Tax brackets & overrides | Configure slabs, override | TaxStatus, TaxOverride |

| ****Tax Adjustment Requests**** | Review tax change requests | Approve, reject | TaxAdjustmentRequest |

| ****Offer Letters**** | Offer letter management | Create, email, track | OfferLetter |

| ****Document Creation**** | Generate HR documents | Select type, generate PDF | Multiple models |

| ****Documents**** | Document library | Upload, assign | EmployeeDoc |

| ****Document Requests**** | Request docs from candidates | Send request link, track | DocumentRequest |

| ****Onboarding**** | Manage new hire onboarding | Send link, track progress, approve | Employee (onboarding status) |

| ****Departments**** | Manage departments | CRUD | Departments |

	Sub-Departments		Manage sub-departments		CRUD		SubDepartments	
	Designations		Manage job titles		CRUD		Designations	
	Shifts		Manage work shifts		CRUD, assign		Shift, EmployeeShift	
	Settings		System configuration		Font, timezone, company settings		Settings, FontSetting	
	Company Profile		Company details		Edit name, logo, address		CompanyProfile	
	HR Policies		Policy management		Write, publish, archive		HrPolicy	
	Penalties		Disciplinary penalties		Issue, track		Penalty	
	Warnings		Formal warnings		Issue, view history		EmployeeWarning	
	Probation		Probation tracking		Create, extend, complete		ProbationPeriod	
	Notice Period		Resignation tracking		Configure, track last day		NoticePeriod	
	Events		Company event calendar		Create, manage		Events	
	Celebrations		Birthday/anniversary view		View upcoming		Employee birthdates	
	Activity Log		System-wide audit trail		View, filter by action/user		Change logs	
	Notifications		Pending items		Approve from notification		Multiple models	
	Permissions Manager		Employee access control		Configure per employee		RolePermission	
	User Management		Admin/HR accounts		Create, edit, 2FA		Users	
	Bugs		Bug tracker		Create, status update		Bug	
	Tasks		Task management		Create, assign, update		Task	
	Payment Requests		Unified financial requests		Bulk review, approve		UnifiedRequests	

6.2 Employee Dashboard — Pages

	Screen		Purpose		Key Actions		Data Source	
	---		---		---		---	

| **Dashboard** | Personal summary | View attendance, leave, announcements | Employee, Attendance, ApplyLeave |

| **Attendance** | Personal attendance | Clock in/out, view history | Attendance, EmployeeSession |

| **Leave Application** | Apply for leave | Multi-step form submission | ApplyLeave |

| **Leave History** | Past leave records | View status, filter | LeaveRecord |

| **Leave Balance** | Remaining balance | View by type | LeaveYearBalance |

| **Salary Slips** | View pay slips | Download PDF, view breakdown | SalarySlip |

| **Loans** | View personal loans | EMI schedule, balance | LoanRequest, LoanDetail |

| **Communications Hub** | Unified inbox | Switch between email/WhatsApp/chat | Multiple |

| **Email Client** | Email management | Read, compose, reply, filter | Email (IMAP) |

| **Team Messages** | Internal threading | Send, react, reply, attach | ThreadChatMessage |

| **Profile** | Personal profile | Edit contact, upload photo | Employee |

| **Profile Revisions** | Change history | View past profile changes | ProfileRevision |

| **My Requests** | All submitted requests | Track status, create new | Multiple request models |

| **Salary Change Request** | Request salary revision | Submit form | SalaryChangeRequest |

| **Loan Application** | Apply for loan | Submit form | LoanRequest |

| **Reimbursement** | Submit expense claim | Fill form, upload receipt | ReimbursementRequest |

| **Bonus Request** | Request bonus | Submit form | BonusRequest |

| **Commission Request** | Request commission | Submit form | CommissionRequest |

| **Advance Salary** | Request advance | Submit form | AdvanceSalaryRequest |

| **Attendance Challenges** | Dispute attendance | Submit challenge, track | AttendanceChallenge |

| **Overtime Requests** | Request overtime | Submit, track | OvertimeRequest |

| **Penalties & Warnings** | View issued actions | View only | Penalty, EmployeeWarning |

- | **Bugs Report** | Report a system bug | Submit form | Bug |
- | **Send Query** | HR inquiry | Submit formal query | (Needs confirmation — model unclear)
- |
- | **Birthdays** | Team celebrations | View upcoming birthdays | Employee birthdates |
- | **Events** | Company calendar | View events | Events |
- | **Notifications** | Alerts and approvals | View, mark read | Socket.IO events |
- | **Settings** | Personal preferences | Notification prefs | Settings |
- | **Manager Dashboard** | Manager-specific view | Team overview, approvals | Employees (team scope) |
- | **Manager Assignments** | Assign work | Assign tasks to team | Task |
- | **Client Tasks** | Client project tracking | View client work | ClientInfo, AssignmentMessage |
- |
- | **My Clients** | Assigned clients | View client list | ClientInfo |
- | **Assignment Messages** | Client communications | View, compose | AssignmentMessage |

7. API Documentation

7.1 Authentication APIs

Method	Endpoint	Purpose	Auth Required	Key Parameters
POST	/api/auth/signup`	Register admin user	None	username, email, password, role
POST	/api/auth/login`	Admin login	None	email, password
POST	/api/auth/verify-otp`	Verify 2FA OTP	Partial (temp token)	otp
POST	/api/auth/setup-2fa`	Enable 2FA	Auth	—

POST	`/api/auth/refresh`	Refresh access token	Refresh token	refreshToken
POST	`/api/auth/logout`	Logout admin	Auth	—
GET	`/api/auth/profile`	Get current user	Auth	—
POST	`/api/emp-auth/login`	Employee login	None	email, password
POST	`/api/emp-auth/heartbeat`	Keep session alive	EmpAuth	sessionId
POST	`/api/emp-auth/reactivate-session`	Reactivate after refresh	EmpAuth	sessionId
POST	`/api/emp-auth/logout`	Employee logout	EmpAuth	—
POST	`/api/emp-auth/refresh`	Refresh employee token	Refresh token	refreshToken

7.2 Employee APIs

Method	Endpoint	Purpose	Auth	Key Parameters
---	---	---	---	---
GET	`/api/employees`	List employees	PayrollAuth	department, status, search
POST	`/api/employees`	Create employee	PayrollAuth	Full employee object
GET	`/api/employees/:id`	Get employee	PayrollAuth	—
PUT	`/api/employees/:id`	Update employee	PayrollAuth	Updated fields
DELETE	`/api/employees/:id`	Soft delete	PayrollAuth	—
GET	`/api/employees/list`	Lightweight list	PayrollAuth	—
GET	`/api/employees/hierarchy`	Org chart data	PayrollAuth	—
GET	`/api/employees/permissions/:id`	Get permissions	PayrollAuth	—
PUT	`/api/employees/permissions/:id`	Update permissions	PayrollAuth	Permission object

7.3 Attendance APIs

Method	Endpoint	Purpose	Auth	Notes
POST	/api/attendance`	Check-in or check-out	AttendanceAuth	Toggles based on current state
GET	/api/attendance`	List attendance records	AttendanceAuth	Supports date range, employee filter
GET	/api/attendance/stats`	Monthly statistics	PayrollAuth	month, year
GET	/api/attendance/employee/:id`	Employee attendance	PayrollAuth	date range
PATCH	/api/attendance/:id/challenge`	Submit challenge	EmpAuth	reason, correct time
GET	/api/attendance/challenges`	List challenges	PayrollAuth	status filter
GET	/api/attendance/audit-logs`	Audit trail	PayrollAuth	date range
GET	/api/attendance/logs/between-shift`	Between-shift logs	PayrollAuth	—
POST	/api/setDate`	Manual correction	PayrollAuth	employeeid, date, status
GET	/api/emp-attendance`	My attendance	EmpAuth	month, year

7.4 Leave APIs

Method	Endpoint	Purpose	Auth	Notes
GET	/api/leaves`	List leave types	PayrollAuth	—
POST	/api/leaves`	Create leave type	PayrollAuth	name, maxDays, carryForward
GET	/api/leaves/balance/:employeeid`	Leave balance	PayrollAuth or EmpAuth	leaveYear
POST	/api/apply-leave`	Apply for leave	EmpAuth	type, dates, reason
GET	/api/emp-leaves`	Employee's leave apps	EmpAuth	status filter
PATCH	/api/emp-leaves/:id/approve`	Approve leave	PayrollAuth	—

PATCH	`/api/emp-leaves/:id/reject`	Reject leave	PayrollAuth	reason
GET	`/api/leave-records`	Leave history	PayrollAuth	employeeId, year
POST	`/api/leave-encashment-requests`	Request encashment	EmpAuth	days, amount
POST	`/api/leave-carry-forward-requests`	Request carryforward	EmpAuth	days

7.5 Payroll APIs

Method	Endpoint	Purpose	Auth	Notes
---	---	---	---	---
GET	`/api/salary-slips`	List slips	PayrollAuth	period, employeeId
POST	`/api/salary-slips/generate`	Generate slips	PayrollAuth	periodId
GET	`/api/salary-slips/:id`	Get slip	PayrollAuth	Encrypted data
GET	`/api/salary-slips/decrypt/:id`	Decrypt slip	PayrollAuth	Requires master key
PATCH	`/api/salary-slips/:id`	Edit slip fields	PayrollAuth	Field updates
POST	`/api/send-slip-email/:id`	Email slip	PayrollAuth	—
GET	`/api/salary-settings`	Get settings	PayrollAuth	—
POST	`/api/salary-settings`	Update settings	PayrollAuth	Settings object
GET	`/api/payroll-periods`	List periods	PayrollAuth	—
POST	`/api/payroll-periods`	Create period	PayrollAuth	start, end dates
PATCH	`/api/payroll-periods/:id/lock`	Lock period	PayrollAuth	—
GET	`/api/salary-structure`	Get structure	PayrollAuth	—
PUT	`/api/salary-structure`	Update structure	PayrollAuth	Component percentages

7.6 Financial Request APIs

Method	Endpoint	Purpose	Auth
--------	----------	---------	------

|---|---|---|---|

POST	`/api/loan-requests`	Apply for loan	EmpAuth
GET	`/api/loan-requests`	List loan requests	PayrollAuth
PATCH	`/api/loan-requests/:id/approve`	Approve loan	PayrollAuth
GET	`/api/loans`	Loan schedules	PayrollAuth
POST	`/api/reimbursement-requests`	Submit reimbursement	EmpAuth
GET	`/api/reimbursement-requests`	List reimbursements	PayrollAuth
PATCH	`/api/reimbursement-requests/:id/approve`	Approve	PayrollAuth
POST	`/api/advance-salary-requests`	Request advance	EmpAuth
PATCH	`/api/advance-salary-requests/:id/approve`	Approve advance	PayrollAuth
POST	`/api/commission-requests`	Submit commission	EmpAuth
GET	`/api/unified-requests`	All request types	PayrollAuth

7.7 Communication APIs

Method	Endpoint	Purpose	Auth
--------	----------	---------	------

|---|---|---|---|

GET	`/api/email-receiver`	Inbox emails	UnifiedAuth
POST	`/api/email-receiver/send`	Send email	UnifiedAuth
GET	`/api/email-receiver/:id`	Email detail	UnifiedAuth
GET	`/api/whatsapp-messages`	WhatsApp inbox	UnifiedAuth
POST	`/api/whatsapp-messages/send`	Send WhatsApp	UnifiedAuth
POST	`/api/chat`	Send chat message	UnifiedAuth
GET	`/api/thread-chat/:threadId`	Thread messages	UnifiedAuth
POST	`/api/assignment-messages`	Create client message	UnifiedAuth
GET	`/api/assignment-messages`	List messages	UnifiedAuth

| PATCH | `/api/assignment-messages/:id/approve` | Approve message | PayrollAuth |

7.8 System APIs

| Method | Endpoint | Purpose | Auth |

|---|---|---|---|

| GET | `/api/health` | Health check | None |

| GET | `/api/departments` | List departments | PayrollAuth |

| POST | `/api/departments` | Create department | PayrollAuth |

| GET | `/api/designations` | List designations | PayrollAuth |

| POST | `/api/designations` | Create designation | PayrollAuth |

| GET | `/api/role` | List roles | PayrollAuth |

| POST | `/api/role` | Create role | PayrollAuth |

| GET | `/api/users` | List users | PayrollAuth |

| POST | `/api/users` | Create user | PayrollAuth |

| GET | `/api/company-profile` | Company details | UnifiedAuth |

| PUT | `/api/company-profile` | Update company | PayrollAuth |

| GET | `/api/decryption-keys` | Get master keys | PayrollAuth |

| GET | `/api/tax` | Tax brackets | PayrollAuth |

| POST | `/api/tax/brackets` | Create tax slab | PayrollAuth |

| POST | `/api/generate/pdf` | Generate document PDF | PayrollAuth |

| GET | `/api/org-hierarchy` | Org chart data | PayrollAuth |

8. Database Documentation

8.1 Core HR Models

`Employees`

****Purpose:**** Primary employee record — the central entity of the entire system.

Field	Type	Purpose
---	---	---
`_id`	ObjectId	Primary key
`firstName`, `lastName`	String	Employee name
`email`	String	Login and contact email
`cnic`	String (encrypted)	National ID (AES-256 encrypted)
`phone`	String	Contact number
`dateOfBirth`	Date	For birthday notifications
`gender`	Enum	male / female / other
`address`	String	Residential address
`joiningDate`	Date	Employment start date
`department`	ObjectId → Departments	Department reference
`subDepartment`	ObjectId → SubDepartments	Sub-department
`designation`	ObjectId → Designations	Current job title
`grossSalary`	Number	Current gross salary
`salaryStructure`	ObjectId → SalaryStructure	Salary component breakdown
`bankName`, `accountNumber`	String	Salary disbursement details
`status`	Enum	active, pending, onboarding, resigned, terminated, in_trash
`shift`	ObjectId → Shift	Assigned shift
`supervisingMode`	Enum	direct / needs_approval

`directManager`	ObjectId → Employees	Reporting manager	
`paidLeaveQuota`	Number	Annual paid leave entitlement	
`probationStatus`	String	Current probation state	
`positionHistory`	Array	Past designations and salary changes	
`trustedDevices`	Array	Allowed devices for login	
`owner`	ObjectId → Users	Multi-tenancy owner reference	
`photo`	String	Profile photo path	
`profileCompleted`	Boolean	Onboarding completion flag	

****Relationships:****

- Has many `Attendance` records
- Has many `SalarySlip` records
- Has many `LeaveRecord` records
- Has many `EmployeeSession` records
- Belongs to `Department`, `Designation`, `Shift`

`Users`

****Purpose:**** Admin and HR accounts that manage the system.

Field	Type	Purpose	
---	---	---	
`_id`	ObjectId	Primary key	
`username`	String	Display name	
`email`	String	Login email	

	`password`		String (hashed)		bcryptjs hash	
	`role`		Enum		admin / hr / super-admin	
	`twoFactorSecret`		String		TOTP secret for 2FA	
	`twoFactorEnabled`		Boolean		2FA toggle	
	`owner`		ObjectId → Users		Multi-tenancy reference	
	`permissionOverrides`		Object		Specific permission exceptions	

`Attendance`

****Purpose:**** One record per employee per day capturing clock-in/out data.

	Field		Type		Purpose	
	---		---		---	
	`employeeId`		ObjectId → Employees		Who	
	`date`		Date		Which day	
	`clockIn`		Date/Time		Check-in timestamp	
	`clockOut`		Date/Time		Check-out timestamp	
	`status`		Enum		present, absent, half_day, late, on_leave, holiday	
	`lateDeduction`		Number		Calculated late penalty amount	
	`challengeStatus`		Enum		none, pending, approved, rejected	
	`challengeId`		ObjectId → AttendanceChallenge		Linked challenge	
	`sessionId`		String		Associated login session	
	`owner`		ObjectId		Tenant reference	

`EmployeeSession`

****Purpose:**** Track active employee login sessions for session security and auto-logout.

Field	Type	Purpose
---	---	---
`employeeId`	ObjectId → Employees	Who
`sessionId`	String	Unique session identifier
`loginTime`	Date	Session start
`lastActivity`	Date	Last heartbeat timestamp
`logoutTime`	Date	Session end
`sessionState`	Enum	active / timeout / logout
`autoLogoutFlag`	Boolean	Was this an automatic logout?
`deviceFingerprint`	String	Browser fingerprint
`ipAddress`	String	Client IP
`userAgent`	String	Browser/OS info

`SalarySlip`

****Purpose:**** Monthly salary record for each employee — encrypted at rest.

Field	Type	Purpose
---	---	---
`employeeId`	ObjectId → Employees	Who
`payrollPeriodId`	ObjectId → PayrollPeriod	Which month

`grossSalary`	Number	Before deductions
`netSalary`	Number	Take-home amount
`encryptedData`	String	AES-256-CBC encrypted full breakdown
`components`	Object	Component amounts (Basic, HRA, etc.)
`deductions`	Object	Tax, PF, loans, late penalties
`workingDays`	Number	Days worked that month
`presentDays`	Number	Actual present days
`leaveDays`	Number	Approved leave days
`isManual`	Boolean	Was this manually created?
`emailSent`	Boolean	Was the slip emailed?
`owner`	ObjectId	Tenant reference

`LeaveYearBalance`

****Purpose:**** Track annual leave quota and remaining balance per employee per leave type.

Field	Type	Purpose
`employeeId`	ObjectId → Employees	Who
`leaveYear`	Number	e.g., 2025
`leaveTypeId`	ObjectId → Leaves	Which leave type
`openingBalance`	Number	Balance at start of year
`totalEntitlement`	Number	Annual entitlement
`usedDays`	Number	Days used
`balance`	Number	Remaining balance

| `carryForward` | Number | Carried from previous year |

`ThreadChatMessage`

****Purpose:**** Individual messages in team discussion threads.

| Field | Type | Purpose |

|---|---|---|

| `threadId` | ObjectId → ChatThread | Parent thread |

| `senderId` | ObjectId | Who sent it (User or Employee) |

| `senderType` | Enum | User / Employee |

| `content` | String | Message text |

| `attachments` | Array | File references |

| `replyTo` | ObjectId → ThreadChatMessage | Message being replied to |

| `reactions` | Array | {emoji, userId} pairs |

| `editHistory` | Array | Previous versions of content |

| `isDeleted` | Boolean | Soft delete flag |

| `readBy` | Array | List of users who read it |

| `createdAt` | Date | Timestamp |

`AssignmentMessage`

****Purpose:**** Client-related communication with a multi-level approval chain.

Field	Type	Purpose
---	---	---
`senderId`	ObjectId	Message composer
`receivers`	Array	Target recipients
`clientId`	ObjectId → ClientInfo	Linked client
`content`	String	Message body
`attachments`	Array	File references
`approvalChain`	Array	[[approverId, decision, timestamp]]
`status`	Enum	pending / approved / disapproved
`resubmissionCount`	Number	How many times resubmitted
`owner`	ObjectId	Tenant reference

8.2 Key Model Relationships

...

Users (Admin/HR)

- └─ owns → Employees (via owner field)
- └─ owns → SalarySlips
- └─ owns → Departments, Designations, Shifts

Employees

- └─ belongs to → Department
- └─ belongs to → Designation
- └─ belongs to → Shift

- |— reports to → Employee (directManager)
- |— has many → Attendance (daily records)
- |— has many → SalarySlip (monthly records)
- |— has many → LeaveYearBalance (per type per year)
- |— has many → LeaveRecord (history)
- |— has many → EmployeeSession (login sessions)
- |— has many → LoanRequest
- |— has many → EmployeeDoc
- └— has many → ProfileRevision

PayrollPeriod

- └— has many → SalarySlip

SalarySlip

- |— belongs to → Employee
- |— belongs to → PayrollPeriod
- └— encrypted by → DecryptionKey

LoanRequest

- └— has one → LoanDetail (EMI schedule)

ChatThread

- └— has many → ThreadChatMessage

AssignmentMessage

└─ belongs to → ClientInfo

└─ has many approval steps

...

9. Reports and Dashboard Documentation

9.1 Admin Dashboard Metrics

Metric	Description	Data Source	Update Frequency
--------	-------------	-------------	------------------

---	---	---	---
-----	-----	-----	-----

Total Active Employees	Count of employees with `status = active`	Employees	Real-time
------------------------	---	-----------	-----------

Monthly Payroll Total	Sum of all net salaries for current period	SalarySlip	On generation
-----------------------	--	------------	---------------

Attendance Rate	% of working days employees were present	Attendance	Daily
-----------------	--	------------	-------

Pending Leaves	Count of `ApplyLeave` with `status = pending`	ApplyLeave	Real-time
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Upcoming Birthdays	Employees with birthday in next 7 days	Employees.dateOfBirth	Daily
--------------------	--	-----------------------	-------

Recent Activity	Latest HR actions	Change logs	Real-time
-----------------	-------------------	-------------	-----------

9.2 Payroll Export

- Excel export of all salary slips for a given payroll period
- Contains: Employee name, ID, Department, Gross, Deductions, Net
- Generated server-side, downloaded by admin

9.3 Attendance Reports

- **Daily View:** Grid of all employees × statuses for a selected date
- **Monthly Summary:** Per-employee count of present/absent/late/half-day days
- **Department Report:** Attendance rate per department

9.4 Leave Reports

- **Calendar View:** Visual month/year calendar with leave blocks
- **Balance Report:** Remaining balance per employee per leave type
- **Usage Report:** How many days each leave type has been used company-wide

9.5 Financial Reports

- **Loan Portfolio:** All active loans with outstanding balance
- **Tax Summary:** Monthly tax deduction per employee
- **PF Report:** Employee and employer contributions

9.6 PDF Generation

Generated documents (via Puppeteer):

- Salary slips (emailed or downloaded)
- Offer letters
- Experience certificates
- Salary certificates
- NDAs
- Employment contracts

10. Automations and Notifications

10.1 Cron Jobs

Job	File	Schedule	Action
---	---	---	---
Session Timeout	`sessionTimeoutCron.js`	Every minute	Auto-logout employees idle > 3 minutes
Leave Sync	`leaveSyncCron.js`	Nightly	Sync leave balances, process carryforwards
Bonus Leave Cron	`bonusLeaveCron.js`	Disabled (commented out)	Bonus leave accrual logic (not active)
Probation Check	Inline in `index.js`	Configurable	Auto-update probation status

10.2 Real-time Socket.IO Notifications

Event Trigger	Recipients Notified	Channel
---	---	---
New leave application submitted	HR, Admin	`join_owner` room
Leave approved or rejected	Employee	`join_employee` room
New assignment message	Approval chain members	`join_assignment_managers`
Message approved/disapproved	Sender, all receivers	Socket.IO broadcast
Salary slip generated	Admin (optional)	Admin room
New chat message	Thread members	`join_thread` room
New WhatsApp message	Assigned users	Socket.IO
Attendance challenge submitted	Admin, HR	Admin room
Challenge resolved	Employee	Employee room
Loan approved/rejected	Employee	Employee room
Profile change submitted	HR, Admin	Admin room

10.3 Email Automations

| Trigger | Email Sent | Via |

|---|---|---|

| Onboarding link created | Onboarding link sent to new hire | Nodemailer SMTP |

| Salary slip generated (if enabled) | PDF slip attached and emailed | `sendSlipEmail` route + Puppeteer |

| Offer letter created | Offer letter PDF emailed to candidate | Nodemailer SMTP |

| Document request created | Secure upload link emailed to candidate | Nodemailer SMTP |

| Password reset | Reset link emailed | Nodemailer SMTP |

10.4 IMAP Watcher (Email Monitor)

****File:**** `backend/src/watcher.js`

A background process that continuously monitors the company's email inbox via IMAP:

- New emails are automatically ingested and stored in the database
- Triggers Socket.IO notification to logged-in users with the email client open
- Supports multiple email accounts
- Processes attachments

10.5 Automatic Deductions in Payroll

These run automatically during payroll generation (not a separate cron):

- Loan EMI deductions (from `LoanDetail`)
- Late arrival deductions (from attendance + `AttendanceConfig`)
- Half-day deductions

- Tax deductions (bracket-based)
- PF contributions
- Leave without pay (unpaid leave deductions)

11. Issues, Risks, and Improvement Suggestions

11.1 Known Bugs

Issue	Location	Severity	Description
Attendance Challenge Not Updating checkIn/checkOut	attendanceController.js	High	When a challenge is approved, the corrected checkIn/checkOut times may not persist to the main Attendance document. This was flagged as a bug fix in git history (ccc6a1d). Verify the patch is complete.
Duplicate CORS Middleware Registration	backend/src/index.js	Medium	The CORS middleware and its error handler are registered twice. The second registration is redundant and could cause unexpected behaviour in edge cases. Remove duplicate.
Double Route Mount	index.js	Low	Both /api/loans and /api/loan map to the same router. Likely unintentional. Standardise to one path and update all clients.
hierarchyController and hierarchyRoute imported but not mounted	index.js	Low	These are imported as "to ensure build" but not used. Clean up unused imports.
bonusLeaveCron commented out	index.js	Low/Medium	Bonus leave accrual logic exists but is disabled. If this was a business requirement, it needs to be investigated and either activated or removed.

| **Attendance Challenge Not Updating checkIn/checkOut** | attendanceController.js — challenge approval logic | High | When a challenge is approved, the corrected checkIn/checkOut times may not persist to the main Attendance document. This was flagged as a bug fix in git history (ccc6a1d). Verify the patch is complete. |

| **Duplicate CORS Middleware Registration** | backend/src/index.js lines 155–172 and 202–219 | Medium | The CORS middleware and its error handler are registered twice. The second registration is redundant and could cause unexpected behaviour in edge cases. Remove duplicate. |

| **Double Route Mount** | index.js lines 267–268 | Low | Both /api/loans and /api/loan map to the same router. Likely unintentional. Standardise to one path and update all clients. |

| **hierarchyController and hierarchyRoute imported but not mounted** | index.js lines 45 and 94 | Low | These are imported as "to ensure build" but not used. Clean up unused imports. |

| **bonusLeaveCron commented out** | index.js line 176 | Low/Medium | Bonus leave accrual logic exists but is disabled. If this was a business requirement, it needs to be investigated and either activated or removed. |

11.2 Security Risks

Risk	Location	Severity	Recommendation
------	----------	----------	----------------

---	---	---	---
-----	-----	-----	-----

Static file CORS	`index.js` line 184: `Access-Control-Allow-Origin: *` for `/uploads`	Medium	Restricting upload file CORS to the allowed origins list would prevent unauthorised access to employee documents and salary PDFs.
----------------------	--	--------	---

`/api/employees` has no auth guard	`index.js` line 225	High	The comment says "leave public if intentional" — but an unauthenticated endpoint that returns all employee data is a significant privacy risk. Add at minimum a `requireAuth` guard.
--	---------------------	------	--

Refresh token strategy	`empAuthController.js`	Medium	Confirm refresh tokens are rotated on use and old tokens invalidated. If not, a stolen refresh token remains valid indefinitely.
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CNIC encryption	`crypto.js`	Low	AES-256-CBC is strong but CBC mode is vulnerable to padding oracle attacks. Consider AES-256-GCM for authenticated encryption.
---------------------	-------------	-----	--

Master decryption key storage	`DecryptionKey` model	High	If the master key is stored in the same database as the encrypted data, the encryption provides limited protection if the DB is compromised. Consider storing keys in a secrets manager (AWS KMS, HashiCorp Vault).
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11.3 Code Quality Issues

Issue	Location	Recommendation
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No centralised error handler	Multiple controllers	Implement a global Express error handler to standardise error responses.
----------------------------------	----------------------	--

Inconsistent route naming	Multiple routes	Mix of camelCase (`loanRequestRoutes`) and kebab-case URLs. Standardise to REST conventions.
-------------------------------	-----------------	--

Controllers doing too much	`employeeController.js`, `attendanceController.js`	Some controllers handle business logic, DB queries, and formatting. Consider a service layer.
--------------------------------	--	---

| ****No input validation middleware**** | API endpoints | No consistent use of a validation library (Joi, Zod, express-validator). Missing validation creates injection risks. |

| ****`Needs confirmation` — Send Query model**** | ``Employee_dashboard`` Send Query page | A "Send Query" screen exists in the employee portal but the corresponding backend model/route is unclear from the codebase. |

| ****WhatsApp API provider unclear**** | ``whatsAppMessageController.js`` | The specific WhatsApp API integration (360Dialog, Twilio, Meta Business API) is not clear. Document the provider and credentials needed. |

11.4 Missing Documentation

| What's Missing | Recommended Action |

|---|---|

| No ``.env.example`` file with all required variables | Create a comprehensive ``.env.example`` listing every env variable with descriptions |

| No API versioning | Add ``/api/v1/`` prefix for future-proofing |

| No Swagger/OpenAPI spec | Generate API documentation using Swagger for developer onboarding |

| Socket.IO event catalogue | Document all event names, payloads, and rooms in a central reference |

| Multi-tenancy implementation | Document how the ``owner`` field creates tenant isolation |

| Decryption key lifecycle | Document how keys are created, rotated, and retired |

| IMAP watcher configuration | Document email server credentials, polling intervals, and failure handling |

11.5 Improvement Suggestions

| Suggestion | Priority | Benefit |

|---|---|---|

| Add rate limiting to auth endpoints | High | Prevent brute force attacks on login |

| Add request logging middleware (Morgan/Winston) | Medium | Better debugging and audit trail |

| Implement database indexing on frequently queried fields | Medium | Performance improvement (index on `employeeId`, `date`, `status`) |

| Add a dedicated notification model | Low | Currently notifications are inferred from Socket.IO; persist them for offline users |

| Implement pagination on all list endpoints | Medium | Performance — currently some endpoints may return all records |

| Add soft-delete to all critical models | Low | Maintain data integrity by never hard-deleting important records |

| Create a unified request status model | Low | Avoid querying 6 different models for the unified requests view |

| Add health check for dependent services | Low | Monitor IMAP, SMTP, and Socket.IO connectivity |

| Set up API documentation process | High | Future documentation: all new routes should include Swagger annotations from day one |

12. Final Summary

This application is a **production-grade, multi-tenant HRMS + ICMS platform** deployed under the Innand/Virsme brand. It comprehensively manages the entire employee lifecycle from recruitment and onboarding through to payroll, leave management, disciplinary actions, and offboarding.

What the System Does Well

- **Comprehensive coverage** — Nearly every HR function is implemented: onboarding, payroll, attendance, leave, loans, reimbursements, warnings, promotions, notice periods, and probation.
- **Strong security foundation** — JWT authentication, 2FA, AES-256 salary encryption, bcrypt password hashing, and session management with device fingerprinting.
- **Real-time architecture** — Socket.IO provides live notifications for approvals, messages, and attendance events without requiring page refresh.
- **Integrated communication** — Email (IMAP/SMTP), WhatsApp, and internal threaded chat in a single platform is a significant differentiator.
- **Audit trails** — Attendance change logs, profile revisions, and session logs provide meaningful auditability.

Critical Items to Address

1. **Security:** The `/api/employees`` endpoint without authentication is the highest-priority issue and should be resolved immediately.
2. **Known attendance challenge bug:** The `checkIn/checkOut` update on challenge approval needs verification and testing.
3. **Duplicate code:** CORS duplicate registration and double route mounts should be cleaned up.
4. **Input validation:** Implement validation middleware across all POST/PUT endpoints to prevent data integrity issues.
5. **Documentation:** Create a `.env.example``, Swagger spec, and Socket.IO event catalogue before onboarding new developers.

Application Scale

| Metric | Count |

|---|---|

Backend route files	80+
MongoDB models	90+
Backend controllers	89
API endpoints	150+ (estimated)
Admin pages	50+
Employee portal pages	30+
Custom middleware	11
Socket.IO events	25+
Cron jobs	3 (1 disabled)
Frontend apps	5

Documentation prepared by code analysis — June 2026. All findings are based on direct source code inspection. Items marked "Needs confirmation" require verification with the development team.